

Fermion Localization Width from Stability and Spectrum

Constraining f_w from First Principles

EDC Project

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Abstract

We analyze the fermion localization width parameter f_w (in units of brane thickness δ) from stability and spectral constraints. We derive a physical window $f_w \in [0.5, 1.2]$ from normalizability, ground-state dominance, and variational bounds. The BVP-tuned value $f_w \approx 0.8$ lies comfortably inside this window, supporting the physical interpretation that $f_w \sim O(1)$ is natural. However, the *specific* value 0.8 (vs 0.7 or 0.9) remains **[Cal]** — determined by the I_4 gate fit, not derived from first principles.

Contents

1	Definition of Fermion Width	2
1.1	In the BVP Framework	2
2	Constraint 1: Normalizability	2
2.1	Bound-State Existence	2
3	Constraint 2: Confinement to Brane	3
3.1	Physical Requirement	3
4	Constraint 3: Ground-State Dominance	3
4.1	Spectral Gap	3
5	Constraint 4: Variational Principle	3
5.1	Energy Functional	3
5.2	Optimal Width	4
6	Combined Constraints	4
6.1	The Allowed Window	4
6.2	Why Larger f_w Affects Overlap	4
7	What Is NOT Derived	5
7.1	Why Specifically 0.8?	5
7.2	What Would Upgrade to [Der]	5
8	Stoplight Verdict	6
9	Conclusion	6

1 Definition of Fermion Width

1.1 In the BVP Framework

Definition 1.1 (Fermion Width Parameter f_w). [\[Der\]](#) The fermion width parameter f_w is defined as:

$$f_w = \frac{\sigma_\psi}{\delta} \quad (1)$$

where σ_ψ is the characteristic width of the fermion mode profile $w(\chi)$ in the extra dimension, and δ is the brane thickness.

Several equivalent definitions of σ_ψ are possible:

1. Second moment (variance):

$$\sigma_\psi^2 = \frac{\int_{-\infty}^{\infty} \chi^2 |w(\chi)|^2 d\chi}{\int_{-\infty}^{\infty} |w(\chi)|^2 d\chi} - \left(\frac{\int_{-\infty}^{\infty} \chi |w|^2 d\chi}{\int_{-\infty}^{\infty} |w|^2 d\chi} \right)^2 \quad (2)$$

2. FWHM (Full Width at Half Maximum):

$$\sigma_\psi \approx \frac{\text{FWHM}}{2\sqrt{2 \ln 2}} \approx 0.425 \times \text{FWHM} \quad (3)$$

3. Exponential decay length: For $w(\chi) \propto e^{-|\chi|/\sigma}$:

$$\sigma_\psi = \sigma \quad (4)$$

In the BVP code, f_w parameterizes the domain-wall mass profile:

$$m(\chi) = m_0 \tanh \left(\frac{\chi - \chi_c}{f_w \delta} \right) \quad (5)$$

2 Constraint 1: Normalizability

2.1 Bound-State Existence

Proposition 2.1 (Normalizability Bound). [\[Der\]](#) For a fermion zero mode to be normalizable on a domain wall:

$$\int_{-\infty}^{\infty} |w(\chi)|^2 d\chi < \infty \quad (6)$$

This requires the mode to decay faster than $|\chi|^{-1/2}$ at large $|\chi|$.

For the domain-wall mass profile (5), the zero mode is:

$$w_L(\chi) \propto \exp \left(- \int_0^\chi m(\chi') d\chi' \right) = \left[\cosh \left(\frac{\chi - \chi_c}{f_w \delta} \right) \right]^{-m_0 f_w \delta} \quad (7)$$

Corollary 2.2 (Width Constraint from Normalizability). [\[Der\]](#) For (7) to be normalizable:

$$m_0 f_w \delta > \frac{1}{2} \quad (8)$$

With $m_0 \sim 1/\delta$ (from dimensional analysis):

$$f_w > \frac{1}{2} \quad (9)$$

This gives our first constraint: $\boxed{f_w > 0.5}$.

3 Constraint 2: Confinement to Brane

3.1 Physical Requirement

Proposition 3.1 (Confinement Bound). *[Dc] For the fermion to be effectively 4-dimensional (not leaking into the bulk), its localization width should not exceed the brane extent:*

$$\sigma_\psi \lesssim L_{\text{brane}} \quad (10)$$

With $L_{\text{brane}} \sim f_w \times \delta$, this gives:

$$f_w \lesssim 2\text{--}3 \quad (11)$$

A stricter bound comes from requiring the mode to be well-localized:

Corollary 3.2 (Strict Localization). *[Dc] For the fermion to be localized within $\pm 2\delta$ of the wall center with 95% probability:*

$$\int_{-2\delta}^{+2\delta} |w|^2 d\chi > 0.95 \quad \Rightarrow \quad f_w \lesssim 1.2 \quad (12)$$

This gives: $\boxed{f_w < 1.2}$ for tight confinement.

4 Constraint 3: Ground-State Dominance

4.1 Spectral Gap

Proposition 4.1 (Spectral Gap Requirement). *[Dc] For the ground-state mode to dominate (excited modes suppressed), we need a sufficient gap between eigenvalues:*

$$\lambda_1 - \lambda_0 \gg T_{\text{eff}} \quad (13)$$

where T_{eff} is an effective temperature scale.

For a potential well of width $f_w \delta$ and depth $V_0 \sim 1/\delta^2$:

$$\lambda_n \sim \frac{n^2}{(f_w \delta)^2} \quad \Rightarrow \quad \Delta\lambda = \lambda_1 - \lambda_0 \sim \frac{3}{(f_w \delta)^2} \quad (14)$$

Corollary 4.2 (Gap-Based Constraint). *[Dc] Requiring $\Delta\lambda > 1/\delta^2$ (gap comparable to ground state energy):*

$$\frac{3}{(f_w)^2} > 1 \quad \Rightarrow \quad f_w < \sqrt{3} \approx 1.7 \quad (15)$$

This is weaker than the localization bound but confirms $f_w \sim O(1)$.

5 Constraint 4: Variational Principle

5.1 Energy Functional

Proposition 5.1 (Variational Lower Bound). *[Der] The ground-state energy satisfies:*

$$E_0 = \min_w \frac{\langle w | H | w \rangle}{\langle w | w \rangle} \quad (16)$$

For a trial function $w(\chi) \propto \exp(-|\chi|/(f_w\delta))$, the energy functional is:

$$E[f_w] = \frac{1}{(f_w\delta)^2} + V_{\text{eff}} \quad (17)$$

where the first term is kinetic and the second is potential energy.

5.2 Optimal Width

Proposition 5.2 (Variational Optimum). [Dc] Minimizing $E[f_w]$ with respect to f_w for a harmonic-like potential:

$$V(\chi) \approx \frac{1}{2}\omega^2\chi^2, \quad \omega \sim \frac{1}{\delta} \quad (18)$$

gives the optimal width:

$$\sigma_{\text{opt}} = \sqrt{\frac{\hbar}{m_{\text{eff}}\omega}} \quad (19)$$

With $m_{\text{eff}} \sim 1/\delta$ and $\omega \sim 1/\delta$:

$$\sigma_{\text{opt}} \sim \delta \quad \Rightarrow \quad f_w^{\text{opt}} \sim 1 \quad (20)$$

The variational principle predicts $f_w \sim O(1)$, consistent with the tuned value 0.8.

6 Combined Constraints

6.1 The Allowed Window

Combining all constraints:

Constraint	Bound	Status	Physical Origin
Normalizability	$f_w > 0.5$	[Der]	Zero mode must be square-integrable
Strict localization	$f_w < 1.2$	[Dc]	95% within $\pm 2\delta$
Confinement	$f_w < 2$	[Dc]	Mode stays within brane
Spectral gap	$f_w < 1.7$	[Dc]	Ground state dominates
Variational	$f_w \sim 1$	[Dc]	Energy minimization

Derived Window

$$f_w \in [0.5, 1.2] \quad (21)$$

The tuned value $f_w = 0.8$ is **inside** this window.

6.2 Why Larger f_w Affects Overlap

Proposition 6.1 (Overlap Sensitivity). [Der] The overlap integral $I_4 = \int w_L^2 w_R^2 w_\phi^2 d\chi$ scales as:

$$I_4 \propto (f_w)^p \times \exp\left(-\frac{d_{\text{LR}}^2}{2\sigma^2}\right) \quad (22)$$

where $p > 0$ (polynomial prefactor) and the exponential depends on d_{LR} .

Larger f_w increases the polynomial prefactor but does NOT affect the exponential suppression (which depends on d_{LR}/σ).

From the sensitivity analysis (OPR-21c):

- f_w elasticity: +1.3 (positive, polynomial)
- d_{LR} elasticity: -6.5 (negative, exponential)

The value $f_w = 0.8$ is where the polynomial prefactor is large enough to give $I_4 \sim 10^{-3}$ GeV while staying within the physical window.

7 What Is NOT Derived

7.1 Why Specifically 0.8?

The constraints give a *window* $[0.5, 1.2]$, not a unique value. The specific value $f_w = 0.8$ comes from:

1. **BVP parameter scan:** Scanning over the window to find where $X_{\text{EDC}} \approx X_{\text{target}}$.
2. **I_4 gate matching:** The value 0.8 gives the right overlap suppression combined with $d_{\text{LR}} = 8\delta$.

Status: The value $f_w = 0.8$ is **[Cal]** (calibrated), not derived.

7.2 What Would Upgrade to [Der]

To derive the specific value, we would need:

1. The exact potential shape $V(\chi)$ from the 5D action
2. The exact fermion mass profile $m(\chi)$ from bulk Yukawa structure
3. Solution of the BVP eigenvalue problem with these exact potentials

Currently, these are modeling choices (Gaussian, tanh, etc.), not derived.

8 Stoplight Verdict

Epistemic Status: **YELLOW** [Dc/Cal]

What is derived [Der]:

- Normalizability requires $f_w > 0.5$
- Overlap scales polynomially with f_w
- Second-moment definition of mode width

What is derived conditional [Dc]:

- Strict localization gives $f_w < 1.2$
- Variational principle predicts $f_w \sim 1$
- Physical window $f_w \in [0.5, 1.2]$

What is calibrated [Cal]:

- Specific value $f_w = 0.8$ from BVP scan
- This value passes I_4 gate with $d_{\text{LR}} = 8\delta$

Upgrade path YELLOW $\rightarrow$ GREEN:

1. Derive $V(\chi)$ and $m(\chi)$ from 5D bulk action
2. Solve exact BVP to get ground-state width
3. Show $\sigma_\psi/\delta \approx 0.8$ emerges from first principles

9 Conclusion

We have established:

1. **Physical window:** $f_w \in [0.5, 1.2]$ from stability and localization constraints.
2. **Tuned value inside window:** $f_w = 0.8$ lies comfortably within this window, supporting its physical plausibility.
3. **Not arbitrary:** The value is constrained by physics, not randomly chosen. But the specific value comes from the I_4 fit.
4. **Polynomial sensitivity:** f_w affects overlap polynomially (elasticity +1.3), unlike d_{LR} which controls exponentially.

Defendable claim: The value $f_w \approx 0.8$ is **physically natural** (within derived bounds) but not **uniquely derived** (still calibrated within the window).